Guidelines for Test 1

Math 535, Fall 2026

Test 1: in class on Monday, 10/26/2026 on Chapters 1, 2, 3. Close book, calculator allowed.

Topics

1. Given a subset of a vector space. Determine whether it forms a subspace. If yes, give the proof; if not, give a counter example.
2. Given a set S of vectors in a vector space. Test whether a vector is in span(S) (or linear combination of vectors in S); Does S span a given subspace?
3. Given a set S of vectors. Determine whether S is linearly independent or linearly dependent. Determine whether S is a basis for the vector space containing it (or spanned by it).
4. Given a set S of linearly independent vector of a vector space V. Extend it to a basis of V by adding some vectors to S.
5. Find a basis and determine the dimension of a vector space. Know the standard basis for F^n, $M_{m\times n}(F), P_n(F),$ where F is any field (mostly, $F = \mathbb{R}$). Other possible spaces: null space of a linear system; V = span(S), range space of a linear transformation, spaces of $(n \times n)$ symmetric matrices, upper triangular matrices, continuous functions, differentiable functions, integrable functions, etc.
6. Find the coordinate matrix $[\mathbf{v}]_{\boldsymbol{B}}$ of a vector **v** relative to a basis $\boldsymbol{B}$ of any given vector space.
7. Given a map T from one vector space **V** to another vector space **W**. Determine if it is linear or not. Provide a proof if yes or give a counterexample if not.
8. Given a linear transformation T from one vector space **V** to another vector space **W**. Evaluate T(**v**) for any vector **v** in V. Find the coordinate matrix of T(**v**) relative to a basis of **W.** Given a linear transformation T: V→ W and bases β, γ for V and W respectively. Find the standard (transformation) matrix $[T]_{\beta}^{\gamma}$.
10. Determine if a linear transformation T is one-to-one, onto, or isomorphism; Find the null space $N(T)$ of T, a basis for $N(T)$, and nullity of T. Find range of T, a basis of range(T), and rank of T.
11. For linear transformations T: $\mathbf{U} \to \mathbf{V}$ and T': $\mathbf{V} \to \mathbf{W}$, and $\mathbf{u} \in \mathbf{U}, \mathbf{v} \in \mathbf{V}$, evaluate $(T \circ T')(\mathbf{v})$ and $(T' \circ T)(\mathbf{u})$. Find the standard matrix for $T, T', T \circ T'$, and $T' \circ T$.
12. Special case: linear transformation problems with linear systems: applying equivalent conditions for A**x** = **b** to be consistent (rank(A) = rank([A| **b**])). Applying the results about linear systems to answer questions such as null space, nullity, range and rank, invertibility, finding bases, and spanning sets.
13. Basic matrix operations ($\pm,\ \times, A^{-1}, A^t$, Row reduction.)
14. Elementary matrices, inverses, and properties.
15. Structure of solutions to a linear system.

Please review Homework 1-5 and Worksheet 1.

Office Hours (Oct. 25): By Canvas zoom: 7-9 pm.

Possible Study Group: Tuesday/Thursday 2:00 – 4:00 pm, Schmitt Hall 340